Lectures 12 & 13

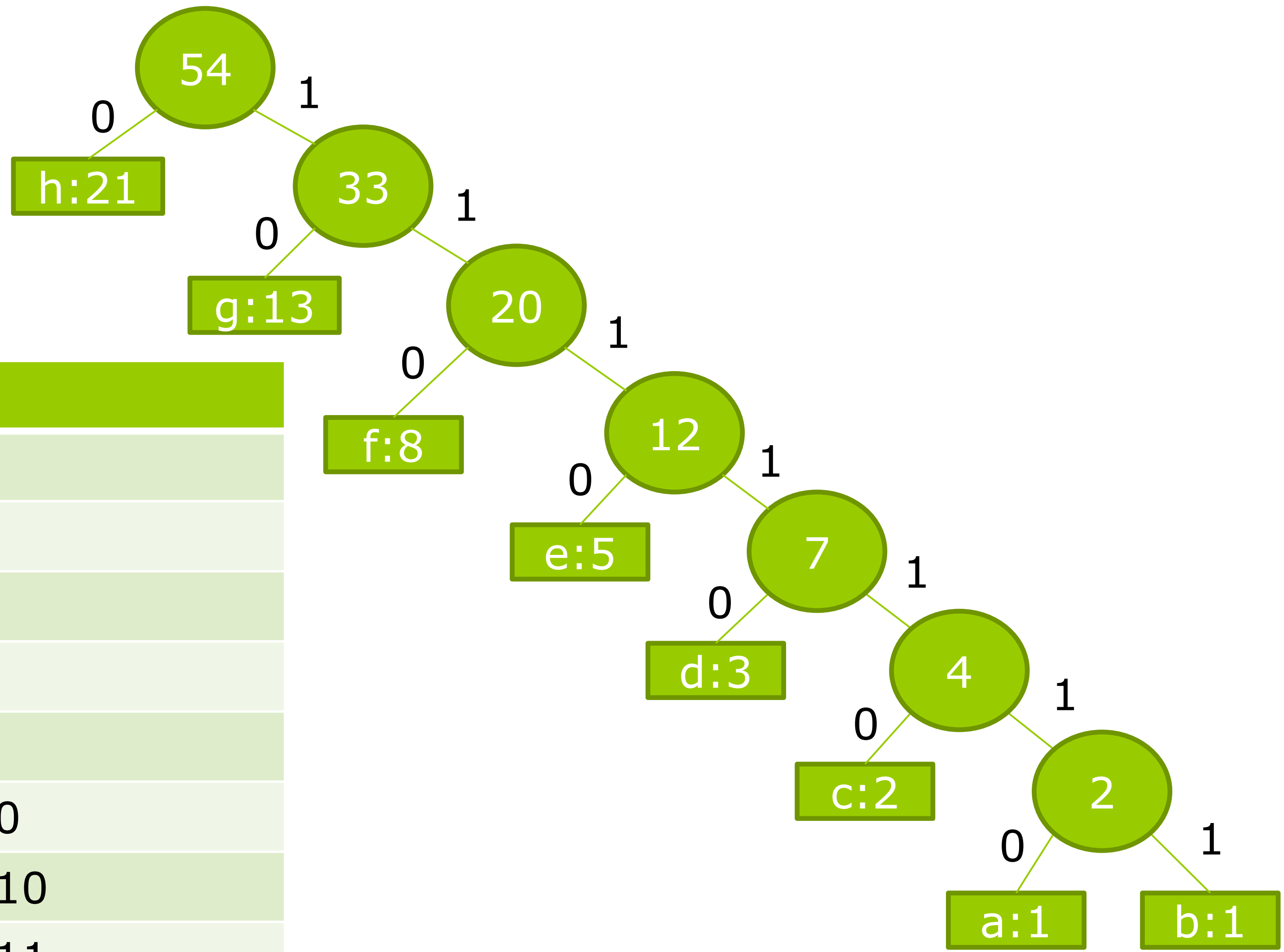
In Class Exercises (solutions)

Exercise 1

What is an optimal Huffman code for the following set of frequencies, based on the first 8 Fibonacci numbers?

`a`:1 `b`:1 `c`:2 `d`:3 `e`:5 `f`:8 `g`:13 `h`:21

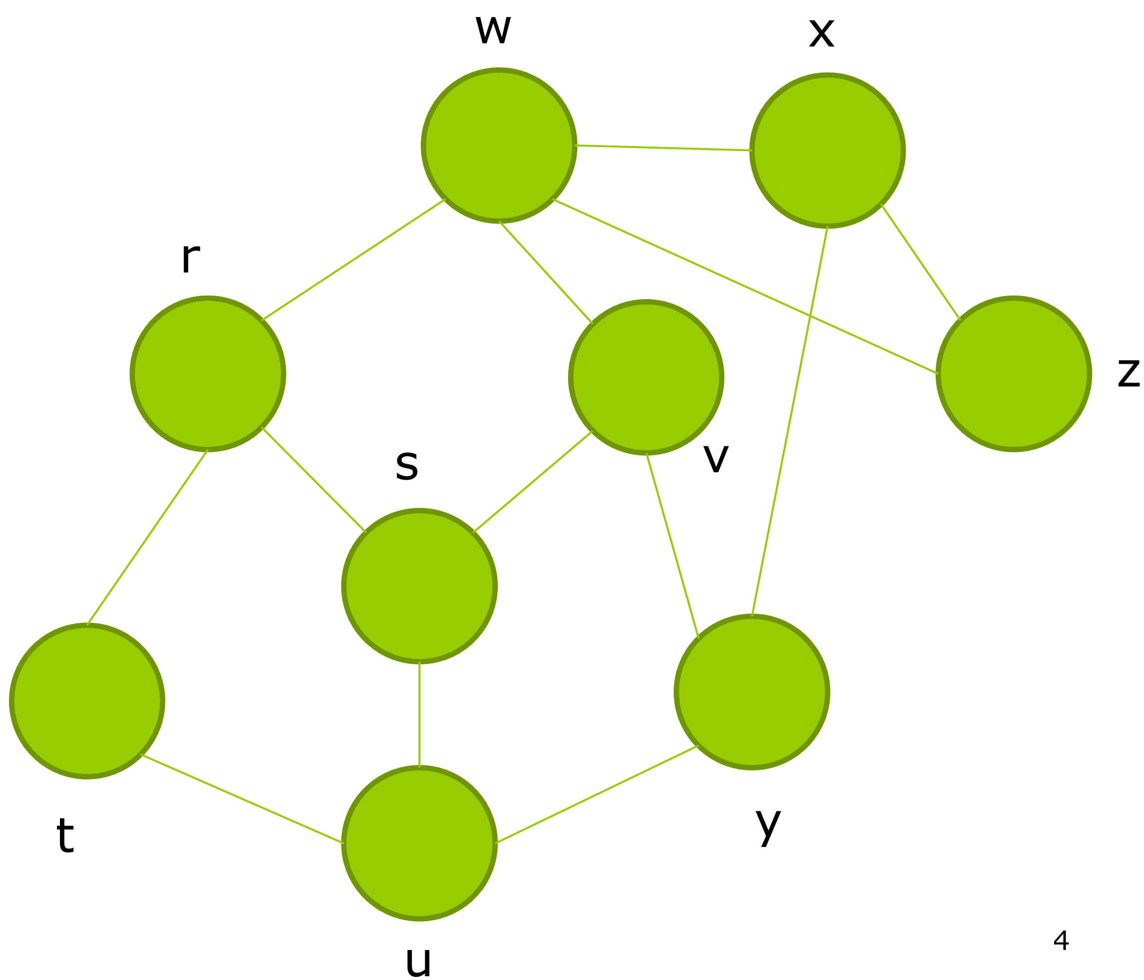
Exercise 1 (solution)



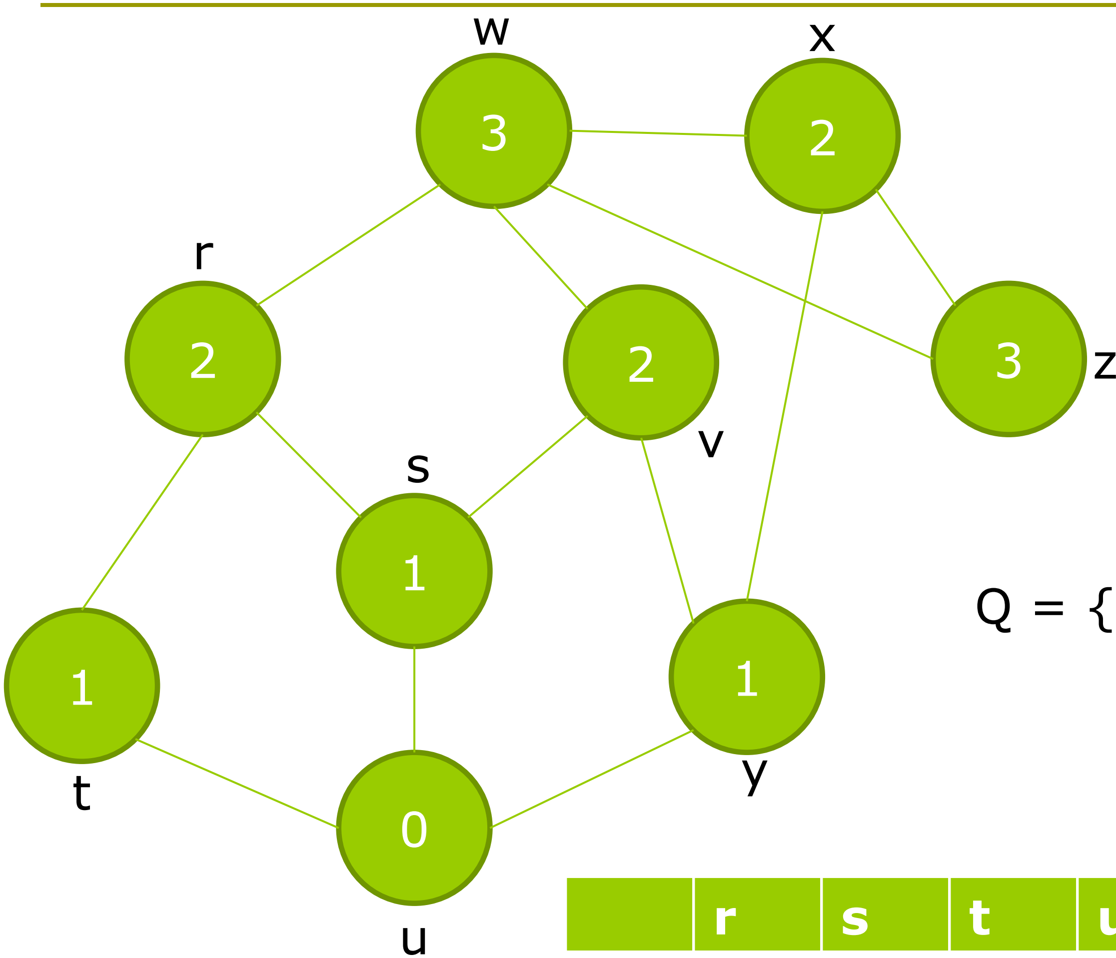
Char	Code
h	0
g	10
f	110
e	1110
d	11110
c	111110
a	1111110
b	1111111

Exercise 2

Show the **d** (distance) and **π** (predecessor) values that result from running Breadth First Search on this undirected graph, using vertex **u** as the source. Assume the neighbors of a vertex are visited in alphabetical order.



Exercise 2 (solution)

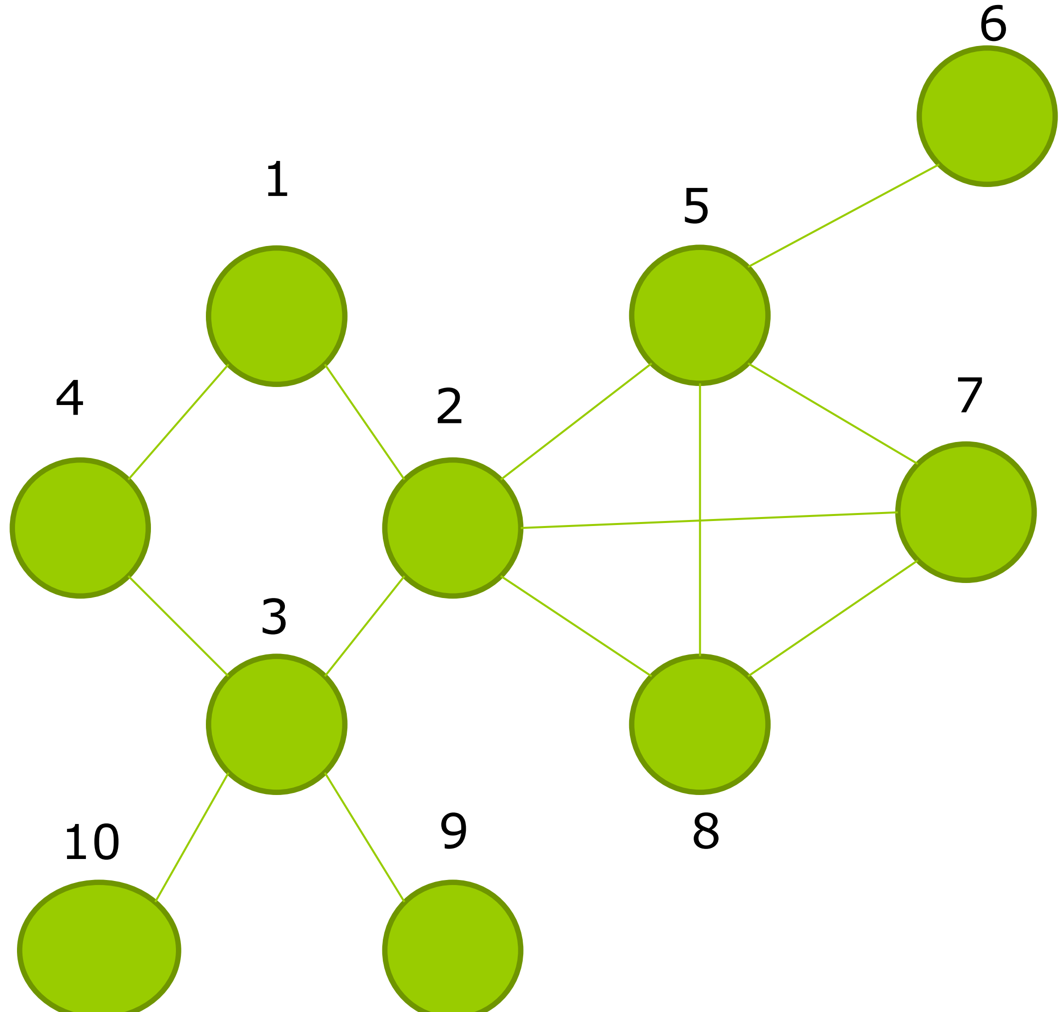


$Q = \{u, s, t, y, r, v, x, w, z\}$

	r	s	t	u	v	w	x	y	z
d	2	1	1	0	2	3	2	1	3
π	s	u	u	Nil	s	r	y	u	x

Exercise 3

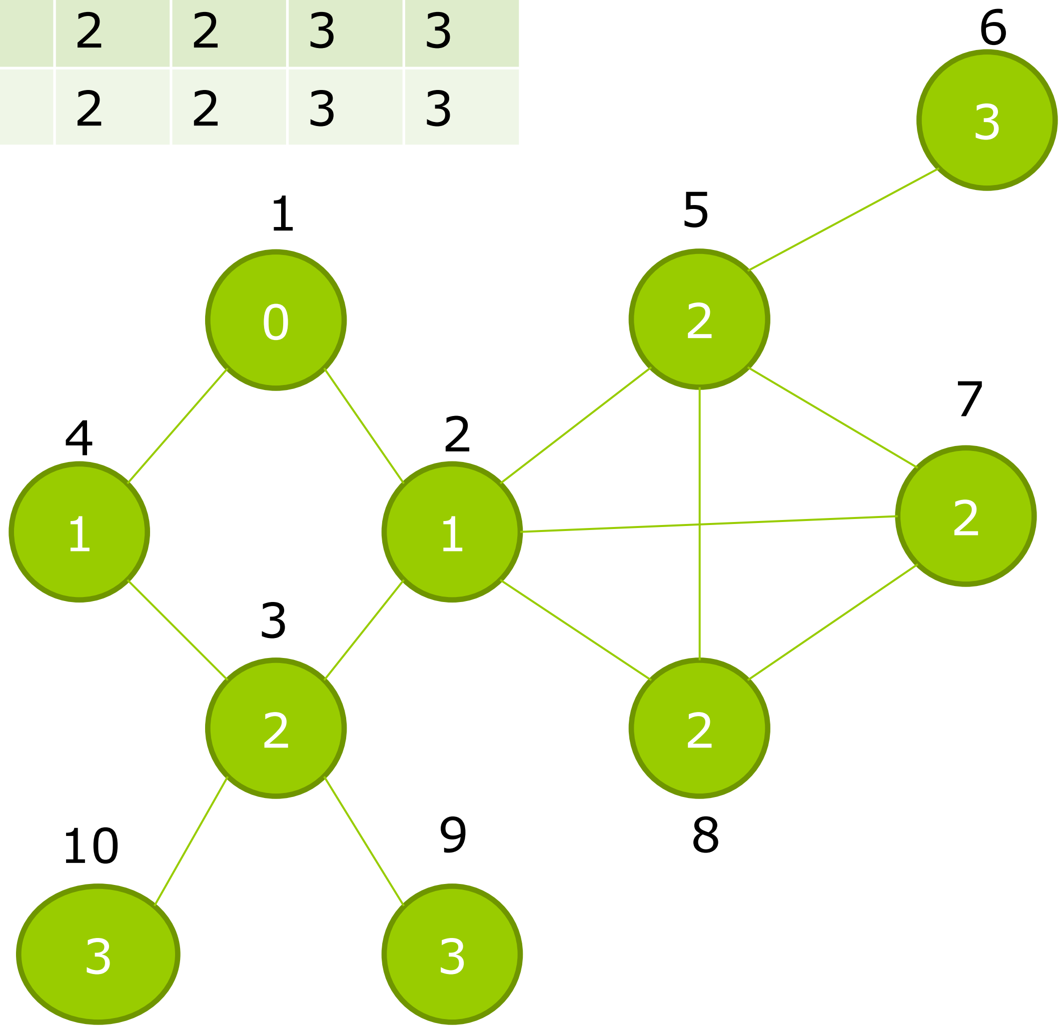
Show the **d** (distance) and **π** (predecessor) values and the queue Q that result from running breadth-first search on the undirected graph of the following figure, using vertex **1** as the source.



Exercise 3 (solution)

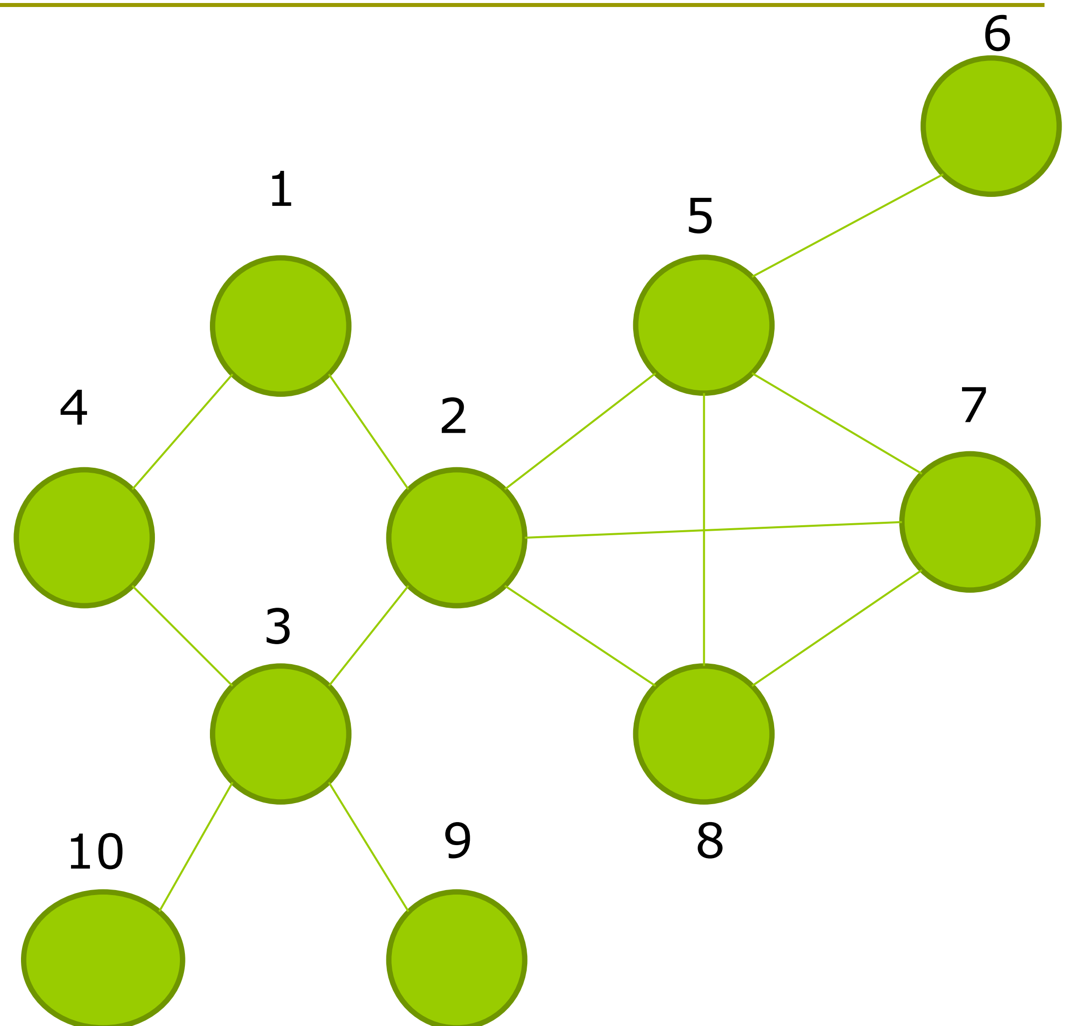
	1	2	3	4	5	6	7	8	9	10
d	0	1	2	1	2	3	2	2	3	3
π	Nil	1	2	1	2	5	2	2	3	3

Q = {1,2,4,3,5,7,8,9,10,6}



Exercise 4

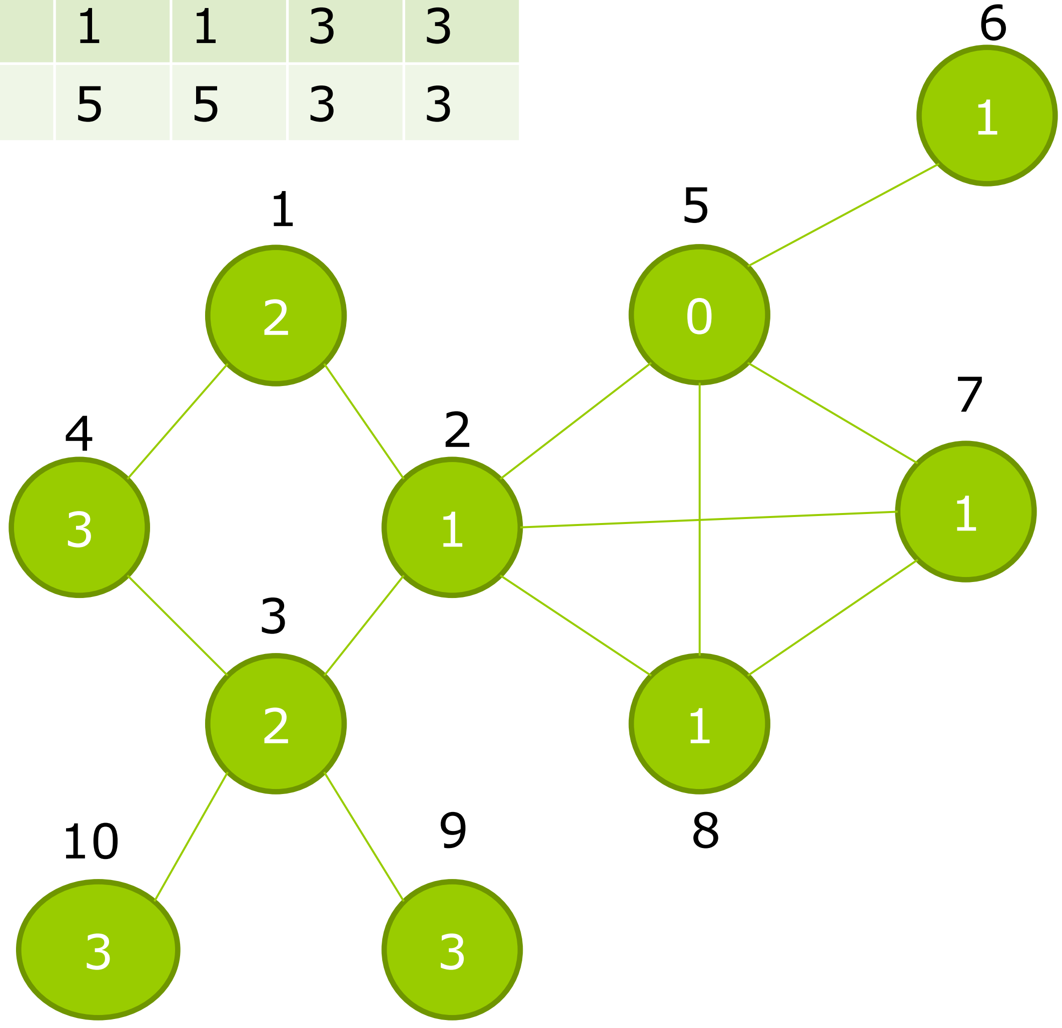
Show the **d** (distance) and **π** (predecessor) values and the queue Q that result from running breadth-first search on the undirected graph of the following figure, using vertex **5** as the source.



Exercise 4 (solution)

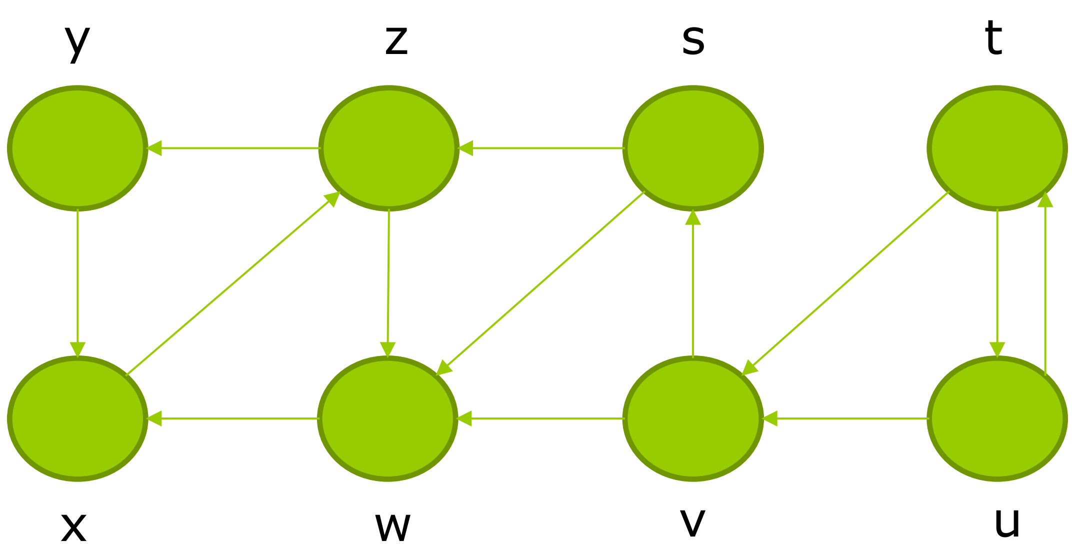
	1	2	3	4	5	6	7	8	9	10
d	2	1	2	3	0	1	1	1	3	3
π	2	5	2	1	Nil	5	5	5	3	3

Q = {5,2,6,7,8,1,3,4,9,10}



Exercise 5

Show how Depth first search works for the following graph. Assume that the DFS procedure considers the vertices in **alphabetical** order and assume that each adjacency list is ordered **alphabetically**. Show the **discovery** and **finish** times for each vertex.



Exercise 5 (solution)

	s	t	u	v	w	x	y	z
d	1	11	12	13	2	3	5	4
f	10	16	15	14	9	8	6	7